

HW/Lab 8: Wireless Sensing: FMCW

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Goals and Instructions

The goals for this assignment are:

- Understand and analyze FMCW radar waveforms and how their parameters control range and velocity performance.
- Implement a MATLAB-based pipeline to estimate range, breathing, and heart rate from simulated radar data.

Follow these instructions:

- **Collaboration:** You can discuss concepts, but write your own solutions and code. **Any plagiarism may result in 0 points.**
- **Tools:** We will primarily use MATLAB. Directions use standard MATLAB functions. *Strongly recommended* toolboxes: Signal Processing, Communications. (Base-MATLAB fallbacks are acceptable.)
- **Submission:** Upload a single PDF with your answers and plots to Gradescope before the deadline. Also, compress (into a ZIP file) and upload your ‘lab8 starter code’ folder to a separate Gradescope portal before the same deadline.
- Total 20 points for ECSE 4560 and 25 points for ECSE 6560 students.

1 Quadratic Chirp FMCW Radar

In class we derived the theory of operation for a linear-chirp FMCW signal. In this problem, we consider a more general *quadratic chirp* and derive its basic properties.

Let the transmitted baseband signal be

$$x(t) = \exp\left(j2\pi\left(f_0t + \frac{k}{3}t^3\right)\right),$$

where f_0 is the starting frequency and k is a real constant.

- (a) Find the instantaneous frequency $f(t)$ as a function of time.

- (b) A target at distance d produces a delayed echo of $x(t)$ with time-of-flight

$$\tau = \frac{2d}{c},$$

where c is the speed of light. Assuming a complex reflection coefficient α , write the received signal

$$y(t) = \alpha x(t - \tau)$$

explicitly in terms of f_0 , k , and τ .

- (c) **(ECSE 6560 only)** At the receiver, the signal is mixed (dechirped) by multiplying the received signal $y(t)$ with the complex conjugate of the transmitted signal $x(t)$, i.e.,

$$s(t) = y(t) x^*(t).$$

Show that $s(t)$ can be written in the form

$$s(t) = A(\tau) \exp\left(j2\pi\left(p(\tau)t^2 + q(\tau)t\right)\right),$$

and derive explicit expressions for $A(\tau)$, $p(\tau)$ and $q(\tau)$ as functions of τ .

Question 2: FMCW Radar Design

Consider the FMCW radar whose transmit frequency–time pattern is shown in Fig. 1. Each up-chirp starts at frequency f_{start} and ramps linearly to $f_{\text{start}} + B_{\text{chirp}}$ over a duration T_{chirp} . Chirps repeat every T_{rep} seconds, and a frame consists of M chirps.

The radar operates at

$$f_{\text{start}} = 77 \text{ GHz}, \quad c = 3 \times 10^8 \text{ m/s},$$

with ADC sampling rate

$$f_s = 25 \text{ MHz}.$$

The sensing requirements for max range, range resolution, max velocity, and velocity resolution respectively are:

$$R_{\text{max}} = 150 \text{ m}, \quad \Delta R = 0.3 \text{ m}, \quad v_{\text{max}} = 20 \text{ m/s}, \quad \Delta v = 0.25 \text{ m/s}.$$

Using standard FMCW relations between these specifications and the parameters in Fig. 1, answer:

- Determine the required chirp bandwidth B_{chirp} (in Hz) to meet the range resolution ΔR .
- Using the maximum unambiguous range R_{max} and sampling rate f_s , determine the required chirp duration T_{chirp} (in μs) and slope k (in Hz/s).
- (ECSE 6560 Only)** Using the maximum unambiguous radial velocity v_{max} , determine the required chirp repetition interval T_{rep} (in seconds).
- (ECSE 6560 Only)** Using the velocity resolution requirement Δv , determine the minimum number of chirps M in each frame.

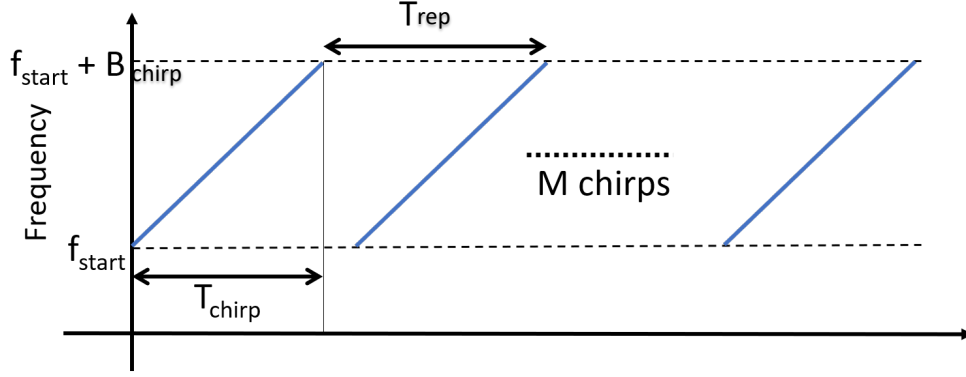


Figure 1: Shows frequency-time plot for FMCW system.

Problem 3: Programming: Breathing and Heart Rate with FMCW Radar

Consider a single-antenna FMCW radar observing a user running on a treadmill. The radar is configured as in Fig. 1, with a linear up-chirp of bandwidth B_{chirp} and duration T_{chirp} repeated M times.

- Start frequency: $f_{\text{start}} = 5 \text{ GHz}$
- Chirp bandwidth: $B_{\text{chirp}} = 1 \text{ GHz}$
- Chirp duration: $T_{\text{chirp}} = 0.1 \text{ } \mu\text{s}$
- Samples per chirp: $N_{\text{sample}} = 256$
- Chirp repetition period: $T_{\text{rep}} = 1/120 \text{ s}$ (slow-time sampling)
- Number of chirps: $N_{\text{chirps}} = 1000$

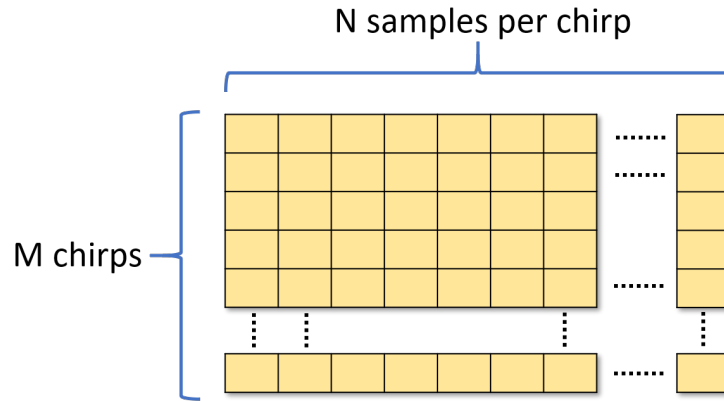


Figure 2: Shows the arrangement of ADC samples in the given matrix.

The user's chest is at an average range $d_0 = 5 \text{ m}$ from the radar and moves due to:

- Breathing: amplitude $d_B = 0.07$ m, rate $f_B = 0.2$ Hz
- Heartbeat: amplitude $d_H = 0.01$ m, rate $f_H = 1.2$ Hz
- Hand motion (micro-Doppler): amplitude 0.1 m, frequency $f_{\text{hand}} = 5$ Hz

At slow-time index m (chirp number), the chest distance is modeled as

$$d_{\text{chest}}[m] = d_0 + d_B \sin(2\pi f_B m T_{\text{rep}}) + d_H \sin(2\pi f_H m T_{\text{rep}}) + 0.1 \sin(2\pi f_{\text{hand}} m T_{\text{rep}}).$$

A second “dancing user” is at an average range of 20 m and moves as

$$d_2[m] = 20 + 0.2 \sin(2\pi f_{\text{dance}} m T_{\text{rep}}), \quad f_{\text{dance}} = 3 \text{ Hz}.$$

This is a simulated model; in real data, motion will be noisier and more irregular. For each target, the round-trip delay is $\tau[m] = 2d[m]/c$, and the received signal is modeled as the sum of delayed FMCW chirps from the chest and the dancing user. The main script (provided `hw8_fmcw_main.m`) simulates the radar Tx and Rx beat signal samples stored in the matrix `adc_sampled` ($N_{\text{chirps}} \times N_{\text{sample}}$).

You are given .p versions of three functions to validate your main script. For submission, you must replace the .p files with your own .m implementations.

Coding tasks

1. **Range FFT and range axis.** Implement the function

$$[\text{RangeFFT}, \text{range_axis}] = \text{compute_range_fft}(\text{adc_sampled}, N_{\text{sample}}, B_{\text{chirp}}, c).$$

The function should:

- Perform an N_{sample} -point FFT along the fast-time dimension for each chirp to obtain `RangeFFT`.
- Construct the corresponding range axis `range_axis` (in meters) using FMCW delay resolution and the relation between delay and range.

2. **Phase extraction in slow time.**

Implement

$$\text{phase_all} = \text{extract_phase}(\text{RangeFFT}, \text{rangeBin}, T_{\text{rep}}).$$

The function should extract and unwrap the phase across chirps at the given range bin.

3. **Breathing/heart-rate spectrum.** Implement

$$[\text{Pabs}, \text{f_axis}] = \text{analyze_fft_for_breathing_heart_rate}(T_{\text{rep}}, N_{\text{chirps}}, \text{phase_all}).$$

The function should:

- Remove the DC component from `phase_all`.
- Compute the magnitude spectrum versus frequency.
- Return the frequency axis `f_axis` (in Hz) and the corresponding magnitude `Pabs`.

Theoretical and interpretation questions: Answer the following questions in your write-up. Show formulas and numerical values where requested.

- (a) Using the given FMCW parameters, derive:
 - (i) The range resolution ΔR and compute its numerical value.
 - (ii) The maximum unambiguous range $R_{\max}$ you can observe with the chosen FFT size.
- (b) Intuitively explain how the breathing and heart rate of the chest target can be estimated using FMCW signal. No formula needed.
- (c) Using your plots:
 - (i) Report the estimated range (in meters) of the chest target.
 - (ii) Report the estimated breathing rate and heart rate (in Hz), as obtained from the slow-time phase spectrum.
- (d) Discuss the impact of additional motions:
 - (i) What is the effect of the dancing user on the range profile and on the range-time map?
 - (ii) How does high-frequency limb/hand motion (micro-Doppler) affect the breathing/heart-rate estimation?

Relate your answers qualitatively to the simulation results.

Deliverables:

- Your completed MATLAB code (with the three `.m` helper functions implemented).
- The plots produced by the script in your report.
- A short PDF write-up with answers to above theoretical and interpretation questions.